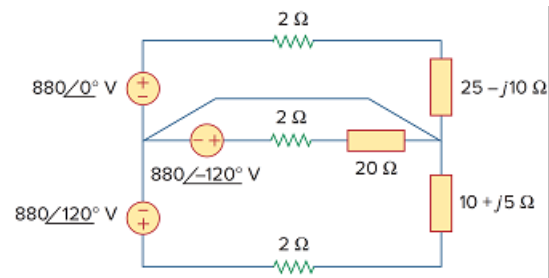


Problem

For the circuit in Fig. 12.43, determine the current in the neutral line.

Figure 12.43

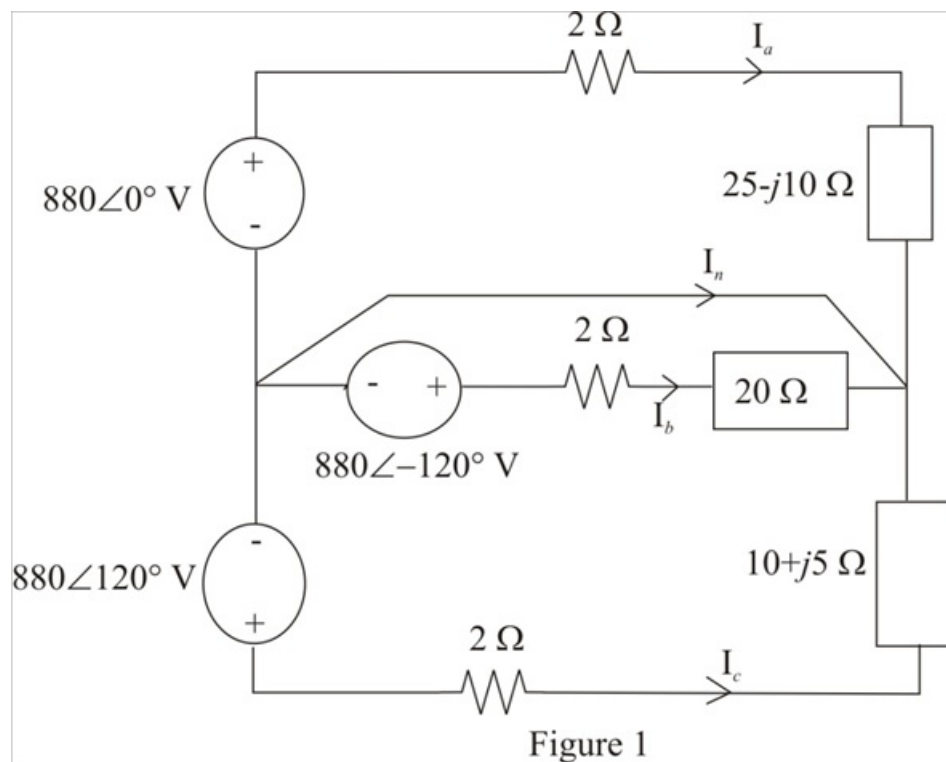


Step-by-step solution

Step 1 of 5

Refer to Figure 12.43 in the text book.

Consider the circuit diagram shown in Figure 1.



Step 2 of 5

Calculate the value of the current line current I_a .

$$\begin{aligned} I_a &= \frac{880\angle 0^\circ}{2+25-j10} \\ &= \frac{880\angle 0^\circ}{27-j10} \\ &= \frac{880\angle 0^\circ}{28.79\angle -20.32^\circ} \\ &= 30.56\angle 20.32^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_a is, $30.56\angle 20.32^\circ \text{ A}$.

Step 3 of 5

Calculate the value of the line current I_b .

$$\begin{aligned} I_b &= \frac{880\angle -120^\circ}{20+2} \\ &= \frac{880\angle -120^\circ}{22} \\ &= 40\angle -120^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_b is, $40\angle -120^\circ \text{ A}$.

Step 4 of 5

Calculate the value of the line current I_c .

$$\begin{aligned} I_c &= \frac{880\angle 120^\circ}{2+10+j5} \\ &= \frac{880\angle 120^\circ}{12+j5} \\ &= \frac{880\angle 120^\circ}{13\angle 22.62^\circ} \\ &= 67.69\angle 97.39^\circ \text{ A} \end{aligned}$$

Therefore, the line current I_c is, $67.69\angle 97.39^\circ \text{ A}$.

Step 5 of 5

Calculate the value of the current I_n in the neutral wire.

$$I_n = -(I_a + I_b + I_c)$$

Substitute $67.69\angle 97.39^\circ \text{ A}$ for I_c , $40\angle -120^\circ \text{ A}$ for I_b and $30.56\angle 20.32^\circ \text{ A}$ for I_a .

$$\begin{aligned} I_n &= -(30.56\angle 20.32^\circ + 40\angle -120^\circ + 67.69\angle 97.39^\circ) \text{ A} \\ &= 43.099\angle 90.06^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_n in the neutral wire is $43.099\angle 90.06^\circ \text{ A}$.